Transcript: Baryonic Suppression of the Matter Power Spectrum: A Literature-Based Survey Without an Emulator Evaluation

HEP-Genesis-Agent session record
Project: ramp-research-baryonic-pk

2026-10-17

The conversation verbatim, generated mechanically from the session record.

A Conversation transcript

The exchange verbatim (assistant text as shown to the user; tool calls summarized by name — see the tool-call log for arguments and results).

User (2026-10-17 11:00:39)

Survey recent work on baryonic suppression of the matter power spectrum. Use the hydrodynamics emulator tools to predict the Pk-ratio at z=0 for a couple of parameter points, explain the AGN feedback physics of suppression, and assess the impact on weak-lensing cosmology.

Assistant (research mode, 2026-10-17 11:00:39)

Plan:

1. (closed) Obtain the matter power spectrum ratio S(k) = P_hydro/P_DMO at z=0 over k = 0.1-10 h/Mpc for at least two feedback settings, a weak/fiducial one and a strong-ejection one. For each, report the depth of the minimum, the k at which the minimum occurs, and the k at which suppression first reaches 1%.
2. (closed) Explain the AGN feedback physics behind the suppression and bracket the predictions from different simulations. The explanation should cover gas ejection from group-scale haloes, the halo-mass dependence, the relation to baryon fraction, the effect of cooling at small scales, and the weak-versus-strong spread between simulations.
3. (closed) Assess the impact on weak-lensing cosmology. Cover the fractional change in cosmic-shear C_ell at ell~500-5000, the S8 bias that results if feedback is ignored, whether feedback can resolve the S8 tension, and constraints from joint WL+SZ analyses.
4. (blocked) Stand in for the missing emulator evaluation. Read numerical S(k, z=0) = P_hydro/P_DMO curves at two clearly defined parameter points from published emulator figures: a weak point (low gas ejection) and a strong point (high ejection). For each point, report S_min, k_min, k(1%), S(k=1) and S(k=5), together with the stated parameter values (BCemu log Mc, or SP(k) f_b at M500~1e14).

5. (blocked) Fix the contested evidence and fill the strong end of the simulation bracket. Obtain verbatim quotes for the key numbers in 1104.1174 (OWLS), 2410.17109 (FLAMINGO jet and low-fgas variants, S(k=1) and S_min) and 2605.04176 (Astrid). Use these to build a weak-to-strong bracket of S(k=1) and S_min at z=0.

6. (closed) Give the weak-lensing assessment quoted numbers. Obtain verbatim support for A_mod~0.69 (KiDS) and the DES Y3 A_mod best fit from 2206.11794 and 2305.09827. Find any quoted suppression at k=0.2-1 h/Mpc, any fractional change in C_ell, or any S8 shift in sigma attributable to ignoring baryons, and quote the FLAMINGO 2309.07959 conclusion that feedback cannot resolve the tension.

Tool calls: `list_agent_files`, `list_input_files`, `list_datasets`, `list_files`, `get_dispatch`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv_abstracts`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `search_arxiv`, `list_input_files`, `list_agent_files`, `get_dispatch`, `run_test_job`, `get_paper_figures`, `download_arxiv_full_text`, `download_arxiv_full_text`, `download_arxiv_full_text`, `download_arxiv_fu` `download_arxiv_full_text`, `download_arxiv_full_text`, `download_arxiv_full_text`, `download_arxiv_fu` `download_arxiv_full_text`, `download_arxiv_full_text`, `download_arxiv_full_text`, `read_arxiv_chunk`, `download_arxiv_full_text`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk` `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `post_dead_end`, `post_finding`, `post_finding`, `read_arxiv_chunk`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `cite`, `cite`, `cite`, `cite`, `cite`, `post_finding`, `cite`, `get_arxiv_metadata`, `get_arxiv_metadata`, `get_arxiv_metadata`, `cite`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `post_question`, `list_files`, `list_files`, `download_arxiv_fu` `read_arxiv_chunk`, `read_arxiv_chunk`, `get_paper_figures`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `get_paper_figures`, `list_files`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `read_arxiv_chunk`, `post_finding`, `post_finding`, `post_dead_end`, `post_finding`, `post_finding`, `post_dead_end`, `post_finding`, `post_finding`, `post_finding`, `post_finding`, `post_dead_end`

Survey — 29 tools on 2 servers, 5 read-only probes, 19 papers found.

Note first: I had no hydrodynamics emulator or Pk-ratio tool in this session. My tools were literature search only, so I did not compute P_hydro/P_DMO at z=0 for any parameter point. That has to be done downstream with an emulator the team actually has, such as BCemu, BACCO, SP(k)/pyspk or the FLAMINGO GP emulator. The search_arxiv_abstracts tool returned HTTP 429 (rate-limited) on all three calls, so everything below comes from search_arxiv and covers abstracts only. The literature broadly agrees that AGN feedback is the main driver of suppression. It pushes gas out of group-scale haloes (M ~ 1e13-1e14 Msun/h), from inside r500 to beyond r200m. This lowers total matter power from about 1% at k~0.3 h/Mpc to about 10-30% near k~1-10 h/Mpc at z=0. Radiative cooling raises power only at very small scales, about k>70 h/Mpc in OWLS. Across simulations, the size of the suppression tracks the mean baryon (or gas) fraction in groups and clusters closely (van Daalen+2019, SP(k)). Predictions still differ widely between simulations: Horizon-AGN and Astrid show weak or almost no suppression at z=0, while FLAMINGO jet models, BAHAMAS and the DES Y3 weak-lensing + ACT kSZ fits show strong suppression. For weak lensing, FLAMINGO finds that realistic feedback is too weak to remove the S8 tension. Amon and Efstathiou and Preston et al. find that resolving it would need suppression stronger than most simulations give, reaching down to k~0.2 h/Mpc. The DES Y3 + kSZ joint fit does prefer such a more extreme feedback scenario, with S8=0.823+0.019-0.020.

Quantities worth checking: Pk-ratio S(k)=P_hydro/P_DMO at z=0 over k=0.1-10 h/Mpc for at least two feedback settings (e.g. weak versus strong ejection): the depth of the minimum,

the k where it occurs, and the k where suppression first reaches 1%; Mean baryon fraction f_b(M500 ~ 1e14 Msun) / (Omega_b/Omega_m) for each point, checked against the SP(k) / van Daalen+2019 relation; BCemu/BACCO parameters: log Mc (gas-retention mass scale), theta_ej (ejection radius), eta, beta; how the suppression responds to Mc; Maximum suppression near k~7 h/Mpc compared with the 20-28% bound from X-ray gas fractions (Giri and Schneider); Suppression at k~0.2-1 h/Mpc compared with what the Amon and Efstathiou / Preston et al. S8 solution requires; Implied fractional change in the cosmic-shear C_ell (ell~500-5000) and the resulting shift in S8 if feedback is ignored (shift of order 1 sigma for DES/KiDS).

Plan — 3 lines of investigation:

1. Obtain the matter power spectrum ratio S(k) = P_hydro/P_DMO at z=0 over k = 0.1-10 h/Mpc for at least two feedback settings, a weak/fiducial one and a strong-ejection one. For each, report the depth of the minimum, the k at which the minimum occurs, and the k at which suppression first reaches 1%. — This answers the task's request for a prediction most directly. The tool catalog has no hydrodynamics or baryonification emulator (no BCemu, BACCO, SP(k) or FLAMINGO tool) and no coding-task tool, so the worker must first confirm that no emulator capability exists. If none exists, the fallback is to take z=0 S(k) curves for named parameter points from emulator and simulation papers, using their figures. These must be labelled as published values, not new emulator predictions.

2. Explain the AGN feedback physics behind the suppression and bracket the predictions from different simulations. The explanation should cover gas ejection from group-scale haloes, the halo-mass dependence, the relation to baryon fraction, the effect of cooling at small scales, and the weak-versus-strong spread between simulations. — The task asks for an explanation of the physics. The survey gives a summary based on abstracts only, which needs grounding in verbatim quotes from full texts. The spread between simulations (Horizon-AGN and Astrid versus BAHAMAS and FLAMINGO jets) also sets the bracket that d-01's parameter points should sit inside.

3. Assess the impact on weak-lensing cosmology. Cover the fractional change in cosmic-shear C_ell at ell~500-5000, the S8 bias that results if feedback is ignored, whether feedback can resolve the S8 tension, and constraints from joint WL+SZ analyses. — The task asks for an assessment of the weak-lensing impact. The relevant results (Amon & Efstathiou, Preston et al., FLAMINGO S8, DES Y3+kSZ, KiDS+tSZ) are all published. They can be compared with the S(k) values from d-01 to show which feedback strength each one requires. Critic's view: revise — No emulator exists in the catalog, so this direction cannot produce the 'prediction' the task asks for. The final report must say plainly that S(k) was not computed and that the values below are published values. Keep the dead-end statement as a required deliverable, not an optional step.; Drop run_test_job. It draws random samples and cannot evaluate P_hydro/P_DMO. list_input_files and list_datasets already show an empty session, and get_dispatch shows local execution. One call to list_input_files is enough to document the absence.; get_paper_figures returns images and captions, not numeric data. Values read by eye for S_min, k_min and especially k(1%) are not verifiable. Every number in the table must come from a verbatim quote found with read_arxiv_chunk. Readings taken from a figure must be marked 'approximate, read from Fig. N', and k(1%) should be reported only where a paper states it. Otherwise list it as 'not quoted'..

Wave 1 — now exploring, in parallel: Obtain the matter power spectrum ratio S(k) = P_hydro/P_DMO at z=0 over k = 0.1-10 h/Mpc for at least two feedback settings, a weak/-fiducial one and a strong-ejection one. For each, report the depth of the minimum, the k at which the minimum occurs, and the k at which suppression first reaches 1%.; Explain the AGN feedback physics behind the suppression and bracket the predictions from different simulations.

The explanation should cover gas ejection from group-scale haloes, the halo-mass dependence, the relation to baryon fraction, the effect of cooling at small scales, and the weak-versus-strong spread between simulations.; Assess the impact on weak-lensing cosmology. Cover the fractional change in cosmic-shear C_ell at ell~500-5000, the S8 bias that results if feedback is ignored, whether feedback can resolve the S8 tension, and constraints from joint WL+SZ analyses..

Finding: Literature value (not computed): at z=0 the OWLS AGN model (van Daalen et al. 2011, arXiv:1104.1174) suppresses power relative to DMO by 1% at k~0.3 h/Mpc, 10% at k~1 h/Mpc and 30% at k~10 h/Mpc. That gives k(1%)~0.3 h/Mpc, S(k=1)~0.90 and S(10)~0.70, with cooling dominant only at k>~70 h/Mpc. On this basis it serves as the strong, fiducial-AGN reference point.

Finding: Literature (FLAMINGO, arXiv:2410.17109, not computed): the response gets stronger and its minimum moves to smaller scales as z falls. Lower cluster gas fractions and collimated jets (instead of thermal winds) strengthen the suppression. The strongest FLAMINGO models agree with some WL+external-data analyses but are in strong tension with X-ray cluster gas fractions. The GP emulator is accurate to 1% for k<10 h/Mpc and z<3. No numerical S_min or k_min values were extracted within the budget.

Finding: Amon & Efstathiou (arXiv:2206.11794) model the non-linear power as P = P_L + A_mod(P_NL - P_L). With Planck LCDM fixed, a fit to KiDS-1000 cosmic shear needs A_mod ~ 0.69, which is a suppression stronger than recent hydro simulations give. The tension being addressed is KiDS-1000 shear S8 = 0.759+0.024-0.021 and DES Y3 shear S8 = 0.772+0.018-0.017, against Planck S8 = 0.829+-0.012. That is about 9% (2.4-2.7 sigma) low for KiDS and about 7% (2.3-2.6 sigma) low for DES.

Finding: Preston, Amon & Efstathiou (arXiv:2305.09827) find that DES Y3 cosmic shear needs a milder non-linear suppression than KiDS-1000 to agree with Planck LCDM. It is still more extreme than recent simulations, and it must extend into the mildly non-linear regime, to k ~ 0.2 h/Mpc. They name non-standard dark matter as an alternative cause. The DES best-fit A_mod value was not read in the sections examined.

Finding: FLAMINGO (McCarthy et al., arXiv:2309.07959) compares its full suite, including extreme feedback variants, with cosmic shear, CMB lensing, tSZ and their cross-spectra. It concludes that baryonic effects are not large enough to remove the S8 tension. Current cosmic shear data are only marginally sensitive to baryon physics. Stronger feedback improves the D3A-cosmology fit to KiDS/DES only slightly, and that preference largely goes away once a lower S8 is adopted. CMB lensing agrees well with Planck LCDM. The strongest tension comes from tSZ y-y, then shear x tSZ, then shear-shear.

Finding: Bigwood et al. (arXiv:2404.06098) fit DES Y3 WL + ACT DR5 kSZ with BCEmu baryonification. They get S8 = 0.823+0.019-0.020, higher than WL alone. The fit prefers a mean feedback scenario more extreme than most hydro simulations. At M500 ~ 1e13 Msun the baryon fraction comes out about 1.4 sigma below HSC-XXL X-ray. At M500 ~ 4e13 Msun, FLAMINGO predicts f_b about 1.5 sigma higher, MillenniumTNG about 3.1 sigma higher and FABLE about 1.4 sigma higher; BAHAMAS agrees best. They also state that model flexibility alone can shift S8 and degrade its error.

Finding: Literature (BCemu, Giri and Schneider, arXiv:2108.08863, not computed): fitting X-ray group and cluster gas fractions gives a z=0 suppression above the percent level at k=0.1-0.4 h/Mpc (so k(1%) lies at 0.1-0.4 h/Mpc). The maximum suppression is 20-28% (S_min ~0.72-0.80) at k~7 h/Mpc, 68% CL. The OWLS-AGN value of 30% at k=10 h/Mpc sits just beyond the strong edge of this band. The FLAMINGO strongest jet and low-fgas models are stated to be in strong tension with X-ray gas fractions.

Finding: The abstracts give three more results. KiDS-1000 lensing x tSZ (arXiv:2109.04458, HMx) prefers strong feedback on its own and finds the amplitude sigma8(Om/0.3)^0.2 about

2 sigma low versus Planck; adding cosmic shear tightens S8 by 40% and gives feedback consistent with hydro simulations. DES Y3 harmonic-space shear (arXiv:2203.07128) finds S8 = 0.793+0.038-0.025 (1.5 sigma below Planck), and extending to k_max = 5 h/Mpc constrains feedback while leaving S8 essentially unchanged. OWLS-AGN (arXiv:1105.1075) shows that ignoring baryons biases w0 by about 40% for a Euclid-like survey; a calibrated halo model reduces this to about the statistical error.

Finding: OWLS AGN (van Daalen+2011, arXiv:1104.1174) at z=0: P_hydro/P_DMO falls by 1% at k~0.3, 10% at k~1 and 30% at k~10 h/Mpc; cooling raises power only for k>~70 h/Mpc.

Finding: van Daalen+2019 (arXiv:1906.00968, 92 OWLS/cosmo-OWLS/BAHAMAS spectra): the large-scale suppression comes from feedback heating gas and ejecting it to large distances, which is needed to match X-ray groups. An empirical model that uses only the mean baryon fraction in groups predicts the z=0 suppression to within 1% for k<1 h/Mpc, and differences in group baryon fraction explain the disagreement between simulations.

Finding: van Loon and van Daalen 2023 (arXiv:2309.06479, cosmo-OWLS/BAHAMAS): AGN feedback ejects matter from inside r500c to r200m and possibly beyond in haloes with M500c>~1e13 Msun/h. For k<~3 h/Mpc, group-sized haloes (1e14+-0.25 Msun/h) dominate the power spectrum; at smaller scales the summed contribution of lower-mass haloes takes over. A lower AGN heating temperature damps the effect at k>~4 h/Mpc for M500c>=1e13.25. A parameter-free model built on halo baryon fractions reaches percent-level accuracy down to k~10 h/Mpc.

Finding: Horizon-AGN (Chisari+2018, arXiv:1801.08559) is the weak end of the bracket: suppression at k~10 h/Mpc is <15% at z=0, with an enhancement of a factor of a few at smaller scales from cooling and star formation. Its redshift evolution over z=0-5 is non-monotonic, and the authors conclude its AGN feedback is probably not strong enough.

Finding: Elbers+2024 (FLAMINGO, arXiv:2403.12967): baryonic feedback is coupled to cosmology through f_b/c_v^2 ~ f_b/(Omega_m sigma8)^(1/4), which gives an accurate fitting formula for the suppression. Combined baryonic and non-baryonic suppression (neutrinos, decaying DM) is larger than the sum of its parts.

Completed — Obtain the matter power spectrum ratio S(k) = P_hydro/P_DMO at z=0 over k = 0.1-10 h/Mpc for at least two feedback settings, a weak/fiducial one and a strong-ejection one. For each, report the depth of the minimum, the k at which the minimum occurs, and the k at which suppression first reaches 1%.. This direction was only partly achieved. No emulator was available, so no S(k) values were computed, and the S_min, k_min and k(1%) table is incomplete. All quantities are at z=0, and every value below is read from published papers.

1. Emulator check (dead end x-01): no tool can evaluate P_hydro/P_DMO. The input file list and the out/ directory are both empty (tc-...-17, -18). Dispatch is set to local (tc-...-19). run_test_job only draws random normal samples on the local machine (tc-...-20).

1. OWLS AGN, the strong/fiducial reference (van Daalen et al. 2011, arXiv:1104.1174, abstract quote; finding f-001). Suppression is 1% at k~0.3 h/Mpc, so k(1%)~0.3 h/Mpc. S(k=1)~0.90 and S(k=10)~0.70. The minimum is at or beyond k=10 h/Mpc, so no interior k_min appears in the 0.1-10 h/Mpc range. Cooling raises power only at k>~70 h/Mpc.

1. BCemu fitted to X-ray gas fractions (Giri and Schneider, arXiv:2108.08863, abstract quote; Figure 8 right panel identified but not read; finding f-007). k(1%) falls at 0.1-0.4 h/Mpc. The maximum suppression is 20-28% (S_min ~0.72-0.80) at k_min ~7 h/Mpc, 68% CL.

1. FLAMINGO (arXiv:2410.17109, abstract; finding f-002). Lower cluster gas fractions and collimated jets give stronger suppression. The strongest models agree with some WL+external-data analyses but are in strong tension with X-ray gas fractions. No numerical S_min or k_min values were extracted.

1. Comparison with the Giri and Schneider 20-28% bound near k~7 h/Mpc: OWLS-AGN reaches 30% suppression at k~10 h/Mpc, at or just beyond the strong edge of the band. The strong FLAMINGO jet and low-fgas variants lie on the strong side of the X-ray-inferred band.

Not done:

- No weak-feedback S_min or k_min (for example Horizon-AGN, or a low-Mc BCemu point).
- No S(k=5) values.
- No f_b(M500) values; the SP(k) paper, arXiv:2305.09710, was not read.
- BCemu Figures 2, 3 and 8 are identified but not pixel-read. The files are in figures_2108.08863/ under the run directory.

The full texts of 2108.08863, 2410.17109 and 1104.1174 are saved as .txt files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-baryonic-pk/runs/261017_110039/. Citations for 2108.08863 and 1104.1174 were added to references.bib in the same directory.

Completed — Assess the impact on weak-lensing cosmology. Cover the fractional change in cosmic-shear C_ell at ell~500-5000, the S8 bias that results if feedback is ignored, whether feedback can resolve the S8 tension, and constraints from joint WL+SZ analyses.. Weak-lensing impact assessment (findings f-003 to f-006 and f-008). (1) Amon & Efstathiou (arXiv:2206.11794) model the power as P = P_L + A_mod(P_NL - P_L). Keeping Planck LCDM, KiDS-1000 shear needs A_mod ~ 0.69. The tension is KiDS shear S8 = 0.759+0.024-0.021 and DES Y3 shear S8 = 0.772+0.018-0.017 against Planck S8 = 0.829+-0.012: about 9% (2.4-2.7 sigma) and 7% (2.3-2.6 sigma) low. (2) Preston et al. (arXiv:2305.09827): DES Y3 needs milder suppression than KiDS but still more than the simulations give, and it must reach down to k~0.2 h/Mpc. The DES best-fit A_mod value was not extracted. (3) FLAMINGO (arXiv:2309.07959): even its extreme feedback variants are too weak to remove the tension. Current shear data are only marginally sensitive to baryons, and the slight preference for stronger feedback in the D3A cosmology fades once a lower S8 is adopted. The strongest tension is in tSZ y-y, then shear x tSZ, then shear-shear. (4) DES Y3 WL + ACT kSZ (arXiv:2404.06098): S8 = 0.823+0.019-0.020, with feedback more extreme than most simulations. f_b at M500~1e13 Msun is about 1.4 sigma below X-ray; FLAMINGO, MTNG and FABLE predict f_b 1.5, 3.1 and 1.4 sigma higher at about 4e13 Msun, and BAHAMAS agrees best. (5) KiDS x tSZ (arXiv:2109.04458): the cross-correlation alone prefers strong feedback and is about 2 sigma low; adding shear tightens S8 by 40% and gives feedback consistent with simulations. DES Y3 harmonic space (arXiv:2203.07128): S8 = 0.793+0.038-0.025 at 1.5 sigma, and k_max = 5 h/Mpc leaves S8 essentially unchanged. OWLS-AGN (arXiv:1105.1075): about 40% w0 bias for Euclid if baryons are ignored. Verdict: simulation-calibrated feedback (FLAMINGO) cannot resolve the tension. Resolving it needs A_mod-type suppression reaching k~0.2 h/Mpc, beyond what the simulations give; only the WL+kSZ fit prefers such extreme feedback, and the X-ray gas fractions and FLAMINGO argue against it. Gaps: I did not extract a quoted S8 shift in sigma from ignoring feedback for DES/KiDS, the fractional C_ell change at ell 500-5000,

or the numerical suppression at k=0.2-1 h/Mpc implied by A_mod=0.69. Full texts are in runs/261017_110039/*.txt, and no cite() entries were added.

Completed — Explain the AGN feedback physics behind the suppression and bracket the predictions from different simulations. The explanation should cover gas ejection from group-scale haloes, the halo-mass dependence, the relation to baryon fraction, the effect of cooling at small scales, and the weak-versus-strong spread between simulations.. Partly done. The physical explanation is in place and rests on cited full-text abstract quotes from 5 papers, stored in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-baryonic-pk/runs/261017_110039/references.bib. The bracket table is incomplete: the strong end has no full-text quotes yet, and Astrid was not read. No figures were extracted.

Physics established: (1) Gas ejection: AGN feedback heats gas and pushes it out to large distances, which is needed to match X-ray groups (van Daalen+2019, arXiv:1906.00968). The gas moves from inside r500c to r200m and possibly beyond in haloes with M500c >~ 1e13 Msun/h (van Loon and van Daalen 2023, arXiv:2309.06479). (2) Halo-mass dependence: for k <~ 3 h/Mpc, group-sized haloes of 1e14+-0.25 Msun/h dominate the power spectrum. At smaller scales the summed lower-mass haloes take over. A lower AGN heating temperature damps the effect at k >~ 4 h/Mpc (2309.06479). (3) Baryon fraction: the mean baryon fraction in groups alone predicts the z=0 suppression to within 1% for k < 1 h/Mpc. Differences in group baryon fraction explain why simulations disagree (1906.00968). A parameter-free model built on halo baryon fractions is accurate to the percent level down to k ~ 10 h/Mpc (2309.06479). (4) Cooling: radiative cooling raises the power only for k >~ 70 h/Mpc in OWLS (arXiv:1104.1174). Horizon-AGN shows an enhancement of a factor of a few at scales smaller than k ~ 10 h/Mpc (arXiv:1801.08559). (5) Cosmology coupling: feedback strength scales with f_b/(Omega_m sigma8)^(1/4), and baryonic plus non-baryonic suppression is larger than the sum of the two (Elbers+2024, arXiv:2403.12967).

Bracket at z=0 (partial):

- Horizon-AGN, weak end: suppression < 15% at k ~ 10 h/Mpc; the authors say its AGN feedback is too weak (1801.08559).
- OWLS AGN, intermediate: about 1% at k ~ 0.3, 10% at k ~ 1 (S(k=1) ~ 0.90) and 30% at k ~ 10 h/Mpc (1104.1174).
- Strong end (FLAMINGO jets, BAHAMAS high-AGN): no S(k=1) or S_min values have been quoted from full text yet.

Board posts: findings f-009 to f-013, and question q-05 listing what remains (Astrid 2605.04176, TNG 2511.10634, FLAMINGO 2410.17109 strong-end numbers).

Critic, wave 1 — continue: 0 findings stand up to the evidence, 13 questioned ('Literature value (not computed): at z=0 the OWLS AGN model (van Daalen et al. 2011, arXiv:': The quoted excerpt from tc-d-01-38 contains only the title, authors and header of arXiv:1104.1174. None of the claimed values (1% at k~0.3, 10% at k~1, 30% at k~10 h/Mpc, cooling at k>~70) appears in the quoted text. The abstract sentence needs to be quoted verbatim. Calling OWLS AGN the 'strong' reference also conflicts with d-02, which places it as intermediate.; 'Literature (FLAMINGO, arXiv:2410.17109, not computed): the response gets stronger and its ': The excerpt from tc-d-01-36 is header-only. None of these claims is shown: response strengthening as z decreases, the minimum moving to smaller scales, stronger suppression with jets and low fgas, tension with X-ray gas fractions, and 1% emulator accuracy.; 'Amon & Efstathiou (arXiv:2206.11794) model the non-linear power as P = P_L + A_mod(P_NL - ': Both excerpts (tc-d-03-33, -35) stop at the abstract heading. A_mod~0.69 and the KiDS/DES/Planck S8

values with their sigma levels are not in the quoted text. The percentage-low and sigma figures also look derived rather than quoted, and their provenance is not stated.; 'Preston, Amon & Efstathiou (arXiv:2305.09827) find that DES Y3 cosmic shear needs a milder': The excerpt from tc-d-03-35 is header-only. The 'milder than KiDS' result, the extension to k~0.2 h/Mpc and the non-standard dark matter alternative are not shown.).

Issues: The core task was to use emulator tools to predict S(k) at z=0 for two parameter points, and this was not done. Every number is a literature read, and even those lack verbatim support. The task is unanswered on its primary quantitative deliverable.; Feedback strength is labelled inconsistently. d-01 calls OWLS AGN the strong/fiducial reference, while d-02 calls it intermediate. The weak/strong pair (q-02) is still undefined, so the S_min/k_min table has no clear rows.; The strong end of the bracket is empty: there are no FLAMINGO jet, BAHAMAS high-AGN or WL+kSZ S(k=1) or S_min values. Astrid (no suppression at z=0) and TNG are cited only through board summaries and have no findings..

Next steps after wave 1 — One wave and about 18 tool calls remain, so I am running three tightly scoped directions of about 6 calls each, matching the critic's three suggestions. No directions are carried over open, and d-01 to d-03 stay closed because each is superseded by a narrower follow-up. The primary deliverable was emulator S(k) at two parameter points. A true emulator run cannot be done: the catalog has no emulator, run_test_job only samples random numbers and cannot install BCemu or pyspk, and the task does not require a facility run. The first direction therefore reads published emulator outputs at explicitly parameterised weak and strong points. This also answers q-02, and it compares against the X-ray band at matched scale (k~7). The second and third directions repair the systematic header-only evidence problem. They prioritise the numbers the final answer depends on: the OWLS reference with a consistent label, the strong end from FLAMINGO jet and low-fgas, Astrid at the weak end, and the A_mod values and FLAMINGO S8 verdict for weak lensing. Analytic C_ell and A_mod computations are dropped because no compute tool exists; where the papers do not state these numbers, the gaps will be reported.

New lines of investigation: Stand in for the missing emulator evaluation. Read numerical S(k, z=0) = P_hydro/P_DMO curves at two clearly defined parameter points from published emulator figures: a weak point (low gas ejection) and a strong point (high ejection). For each point, report S_min, k_min, k(1%), S(k=1) and S(k=5), together with the stated parameter values (BCemu log Mc, or SP(k) f_b at M500~1e14).; Fix the contested evidence and fill the strong end of the simulation bracket. Obtain verbatim quotes for the key numbers in 1104.1174 (OWLS), 2410.17109 (FLAMINGO jet and low-fgas variants, S(k=1) and S_min) and 2605.04176 (Astrid). Use these to build a weak-to-strong bracket of S(k=1) and S_min at z=0.; Give the weak-lensing assessment quoted numbers. Obtain verbatim support for A_mod~0.69 (KiDS) and the DES Y3 A_mod best fit from 2206.11794 and 2305.09827. Find any quoted suppression at k=0.2-1 h/Mpc, any fractional change in C_ell, or any S8 shift in sigma attributable to ignoring baryons, and quote the FLAMINGO 2309.07959 conclusion that feedback cannot resolve the tension..

Wave 2 — now exploring, in parallel: Stand in for the missing emulator evaluation. Read numerical S(k, z=0) = P_hydro/P_DMO curves at two clearly defined parameter points from published emulator figures: a weak point (low gas ejection) and a strong point (high ejection). For each point, report S_min, k_min, k(1%), S(k=1) and S(k=5), together with the stated parameter values (BCemu log Mc, or SP(k) f_b at M500~1e14).; Fix the contested evidence and fill the strong end of the simulation bracket. Obtain verbatim quotes for the key numbers in 1104.1174 (OWLS), 2410.17109 (FLAMINGO jet and low-fgas variants, S(k=1) and S_min) and 2605.04176 (Astrid). Use these to build a weak-to-strong bracket of S(k=1) and S_min at z=0.; Give the weak-lensing assessment quoted numbers. Obtain verbatim support for A_mod~0.69 (KiDS) and the DES Y3 A_mod best fit from 2206.11794 and 2305.09827. Find

any quoted suppression at k=0.2-1 h/Mpc, any fractional change in C_ell, or any S8 shift in sigma attributable to ignoring baryons, and quote the FLAMINGO 2309.07959 conclusion that feedback cannot resolve the tension..

Finding: Verbatim from Amon & Efstathiou (arXiv:2206.11794, Eq. 7): with the cosmology fixed to Planck TTTEEE LCDM, the KiDS-1000 xi_pm fit gives "a mean Amod = 0.69 ± 0.06" with chi2_min = 265.5 for 225 data points. The authors note that this uncertainty is underestimated because it neglects the Planck cosmology error. HMCode2020 with a broad prior instead gives Theta_AGN = 9.17+0.2-0.8, well outside the BAHAMAS-calibrated range 7.6-8.0. The paper states that "the power spectrum suppression required to reconcile the KiDS cosmic shear measurements with the Planck cosmology is more extreme than predicted by most of the hydrodynamic simulations" (Fig. 6). For KiDS, ignoring feedback (HMCode2020 no feedback) gives S8 = 0.755+-0.016, against 0.774+-0.021 for the fiducial Theta_AGN = 7.3-8.3 prior (Table 1): a shift of about 0.02, roughly 1 sigma.

Finding: Verbatim from Preston, Amon & Efstathiou (arXiv:2305.09827). With a Planck prior and no scale cuts, DES Y3 gives "Amod = 0.858 ± 0.052", which differs from unity by about 2.7 sigma (Eq. 3). KiDS reanalysed with the same Planck prior gives Amod = 0.748 ± 0.072, about 1 sigma above the 0.69 of Paper I (Eq. 5). The naive combination is Amod = 0.820 ± 0.042 for DES Y3 + KiDS, meaning suppression is required at about 4 sigma (Eq. 6). Over k <~ 1 h/Mpc, BAHAMAS log10(dT_heat/K) = 8.0 "is well approximated by Amod = 0.9". The required suppression is stronger than that, closer to C-OWLS 8.5/8.7, and must extend to k ~ 0.2 h/Mpc (Fig. 2 right, Fig. 6). On S8 when baryons are ignored: DES Y3 with no scale cuts and no feedback (variant 4) gives S8 = 0.793 ± 0.012, which is 1.8 sigma below Planck TTTEEE 0.828 ± 0.016. With scale cuts (variant 3) it gives 0.805 ± 0.017, 0.9 sigma below. Adding Amod with free cosmology (variant 5) gives 0.831 ± 0.036 (Table 2). Removing the baryon-feedback mitigation therefore lowers DES S8 by about 0.04 relative to the Amod-free fit, which is about 1 sigma of that fit.

Finding: Literature value, not computed (BCemu, Giri and Schneider, arXiv:2108.08863, Sec. 4.2 text): S(k, z=0) inferred from X-ray gas fractions, with the 7-parameter BCemu posterior at log10 Mc ~ 12.6, Planck cosmology and b_hse = 0.26+-0.07. At 68% CL, suppression first exceeds 1% at k~0.1-0.4 h/Mpc and reaches a maximum of 21-28% (S_min ~0.72-0.79) at k_min ~7 h/Mpc; the abstract rounds this to 20-28%. At 99% CL, k(1%) lies between 0.1 and 1.3 h/Mpc and the maximum is 14-38% (S_min ~0.62-0.86) beyond k~7 h/Mpc. The posterior pushes delta low, meaning a slowly falling gas profile with large amounts of gas ejected beyond the virial radius.

Finding: Literature statement (arXiv:2108.08863, Sec. 4.2), comparing simulations with the X-ray-informed BCemu S(k) band at z=0: only cosmo-OWLS 8.5 is fully consistent. cosmo-OWLS 8.0, BAHAMAS and OWLS lie between the 68% and 99% regions. Illustris-TNG and Horizon-AGN fall outside the 99% range because their suppression is shifted to higher k. The BCemu emulator covers log10 Mc in [11,15], theta_ej in [2,8], delta in [3,11], eta in [0.05,0.4] and f_b in [0.10,0.25], for k~0.03-12.5 h/Mpc, with 1-sigma emulation error below 1%.

Finding: OWLS AGN (van Daalen+2011, arXiv:1104.1174), verbatim abstract: the AGN run 'predicts a decrease in power relative to a dark matter only simulation ranging, at z = 0, from 1% at k ≈ 0.3 h Mpc-1 to 10% at k ≈ 1 h Mpc-1 and to 30% at k ≈ 10 h Mpc-1', and cooling 'is the dominant effect only for k ? 70 h Mpc-1'. So S(k=1)~0.90 and S(k=10)~0.70. In the bracket it is labelled INTERMEDIATE: it sits stronger than FLAMINGO fiducial and Horizon-AGN, and below the strongest cosmo-OWLS runs.

Finding: FLAMINGO (Schaller+, arXiv:2410.17109), verbatim text quotes. (1) Fiducial: 'The position of the minimum of the response moves from k = 6 h·Mpc-1 at z = 2 to k = 10 h·Mpc-1 at z = 0', and the departure from unity begins at 'k≈ 0.1 h·Mpc-1 by z = 0'. (2)

Thermal fgas variants: 'the minimum stays in the range k = 8 to 10 h·Mpc-1 with a weak dependence on fgas±Nσ'. Below fgas-8σ the response 'reaches a saturation value'. (3) Jet: 'At fixed value of fgas±Nσ, the emulator predicts a larger response than in the case with purely thermal AGN'. (4) The strong end maps to cosmo-OWLS: 'our models with fgas-8σ using the thermal and jet models closely resemble the COSMO-OWLS models with ΔTAGN = 8.3 and 8.5 respectively'. (5) Only Illustris and cosmo-OWLS-AGN-ΔT8.7 lie beyond the emulator's reach. The text does not quote S(k=1) or S_min as numbers. These values appear only in Figs. 5, 6, 11 and 13, whose y-axes span 0.65/0.75 to 1.10.

Finding: FLAMINGO strong end versus data (arXiv:2410.17109), verbatim: 'both fgas-8σ models (i.e. with thermal AGN and with collimated jet AGN) are within the ranges demanded by the analysis of Bigwood et al. 2024 (with or without added kSZ constraints)', yet they 'display gas fractions that are in strong tension with the gas fractions in groups and clusters inferred from joint X-ray and weak-lensing analyses (Kugel et al. 2023)'. Also: the Amon & Efstathiou (2022) response and, to a lesser extent, the Preston et al. (2023) response 'are out of reach of our emulator and likely of what can be achieved using the FLAMINGO model without altering the sub-grid models'.

Finding: Astrid (Yang+, arXiv:2605.04176), verbatim abstract: 'We find no significant suppression at z = 0 and mild suppression at z = 0.2, weaker than in other state-of-the-art simulations'. In the text: 'At z = 0.2, mild suppression extends to k$\sim$1 h Mpc-1, before the power spectrum ratio rises back to above unity at z = 0'. BH repositioning 'boosts kinetic AGN feedback (e.g., by a factor of 2 at z = 1.5)', and at z=0 SV-Repos lies 'lower than that of SV-DF by $\sim$10% around k = 20 h Mpc-1'. Astrid is therefore the WEAK end: S(k=1, z=0) $\sim$ 1 or above, with no interior S_min below 1 quoted.

Completed — Give the weak-lensing assessment quoted numbers. Obtain verbatim support for A_mod~0.69 (KiDS) and the DES Y3 A_mod best fit from 2206.11794 and 2305.09827. Find any quoted suppression at k=0.2-1 h/Mpc, any fractional change in C_ell, or any S8 shift in sigma attributable to ignoring baryons, and quote the FLAMINGO 2309.07959 conclusion that feedback cannot resolve the tension.. Partly done: the A_mod quotes and the S8 shifts are now verbatim, but there is no fractional C_ell change, no suppression number at k=0.2-1 h/Mpc, no FLAMINGO quote from this round and no cite() entries.

1. KiDS (Amon and Efstathiou, arXiv:2206.11794, Eq. 7; finding f-014): with the cosmology fixed to Planck, "a mean Amod = 0.69 ± 0.06", chi2_min = 265.5 for 225 data points. The authors say this error is underestimated because it leaves out the Planck cosmology error. The HMCode2020 version of the fit needs Theta_AGN = 9.17+0.2-0.8 (Eq. 8), well outside the BAHAMAS-calibrated range of 7.6-8.0. The paper says the required suppression is "more extreme than predicted by most of the hydrodynamic simulations" (Fig. 6). From Table 1, S8 for KiDS is 0.755+-0.016 with feedback switched off and 0.774+-0.021 with the fiducial Theta_AGN 7.3-8.3 prior, a shift of about 0.02 (about 1 sigma).

1. DES Y3 (Preston, Amon and Efstathiou, arXiv:2305.09827; finding f-015): with a Planck prior and no scale cuts, Amod = 0.858 ± 0.052, which is 2.7 sigma from 1 (Eq. 3). KiDS refit with the same Planck prior gives Amod = 0.748 ± 0.072 (Eq. 5). Combining the two gives Amod = 0.820 ± 0.042, so suppression is needed at about 4 sigma (Eq. 6). Over k <$\sim$ 1 h/Mpc, BAHAMAS with heating parameter 8.0 is approximated by Amod = 0.9. The data need more suppression than that, close to the C-OWLS 8.5/8.7 runs, and it must reach k $\sim$ 0.2 h/Mpc (abstract; Fig. 2 right, Fig. 6).

1. S8 shift from ignoring baryons (Table 2 of 2305.09827): with all scales used and no feedback, DES gives S8 = 0.793 ± 0.012, 1.8 sigma below Planck TTTEEE (0.828 ± 0.016). With

scale cuts it gives 0.805 ± 0.017, 0.9 sigma below. With Amod and free cosmology it gives 0.831 ± 0.036. So leaving out the suppression lowers S8 by about 0.04, roughly 1 sigma of the Amod fit.

Not found (dead end x-02): neither paper gives a fractional C_ell or xi_+ change at ell 500-5000, or an S(k) number at k=0.2-1 h/Mpc, in the text I read (2206.11794 characters 0-46000, 2305.09827 characters 0-42000). The suppression appears only in figures (2206.11794 Fig. 6; 2305.09827 Figs. 2, 4, 6), and both papers work with xi_pm rather than C_ell.

FLAMINGO (2309.07959): not re-read in this round. The conclusion that feedback is too weak to remove the S8 tension still rests only on finding f-005.

No cite() entries were added. Texts used: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-baryonic-pk/runs/261017_110039/2206.11794.txt and 2305.09827.txt.

Still open: a verbatim quote from the 2309.07959 conclusions, and cite() entries for 2206.11794, 2305.09827 and 2309.07959.

Blocked — Stand in for the missing emulator evaluation. Read numerical S(k, z=0) = P_hydro/P_DMO curves at two clearly defined parameter points from published emulator figures: a weak point (low gas ejection) and a strong point (high ejection). For each point, report S_min, k_min, k(1%), S(k=1) and S(k=5), together with the stated parameter values (BCemu log Mc, or SP(k) f_b at M500~1e14).. (No emulator can run in this session, and no tool can read or digitise the saved figure images. The S(1), S(5), S_min and k_min values for named low-Mc and high-Mc (or f_b) points therefore cannot be obtained. Next step: run BCemu or pyspk in a coding or HPC pack, or digitise the saved PNGs outside this session.) No emulator was evaluated, and the two-point table was not produced. Every value below is a literature value at z=0, not something I computed.

Why no evaluation was possible: the catalog has no emulator tool, BCemu and pyspk cannot be installed, and run_test_job only draws random samples. Why the figures could not stand in: get_paper_figures only saves PNG files and captions, and no tool here can view or digitise an image. So S(k) values could not be read from the published curves, and the planned plus or minus 0.02 reading uncertainty was never applied (dead end x-03).

What was established (findings f-016 and f-017):

1. BCemu fitted to X-ray gas fractions (Giri and Schneider, arXiv:2108.08863, Sec. 4.2 and Fig. 8; 7-parameter posterior with log10 Mc ~ 12.6, a low delta meaning a lot of gas is pushed past the virial radius, Planck cosmology, and hydrostatic mass bias b_hse = 0.26 +- 0.07):

- 68% CL: k(1%) ~ 0.1-0.4 h/Mpc; S_min ~ 0.72-0.79 (21-28% suppression) at k_min ~ 7 h/Mpc.
- 99% CL: k(1%) ~ 0.1-1.3 h/Mpc; S_min ~ 0.62-0.86 (14-38% suppression) beyond k ~ 7 h/Mpc.

1. Verbatim quotes supporting the 20-28% band at k~7 h/Mpc:

- Paper body: "At the 68 percent confidence level, the baryonic power suppression S is predicted to exceed the percent level between k~0.1 and 0.4 h/Mpc growing to a maximum 21-28 percent at around k~7 h/Mpc."
- Abstract: "reach a maximum of 20-28 percent at around k~7 h/Mpc (68 percent confidence level)".

1. Placing simulations against that band at matched scale (the paper's own statements): only cosmo-OWLS 8.5 is fully consistent. cosmo-OWLS 8.0, BAHAMAS and OWLS lie between the 68% and 99% regions. Illustris-TNG and Horizon-AGN fall outside the 99% region, with their suppression shifted to higher k. These are the paper's practical weak and strong examples, but they are named simulations, not named emulator parameter points.

1. BCemu parameter ranges: log10 Mc in [11, 15], theta_ej in [2, 8], delta in [3, 11], f_b in [0.10, 0.25]. It covers k = 0.03-12.5 h/Mpc with emulation errors below 1% at 1 sigma.

Still open:

- The weak (low-Mc) and strong (high-Mc) rows at named parameters.
- S(k=1) and S(k=5) for any point.
- The SP(k) f_b to S relation at k=1 and 5 h/Mpc. The text gives none of these, and I did not read the SP(k) full text.

Files saved in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-baryonic-pk/runs/261017_1100

- figures_2108.08863/ holds fig02.png, fig03_1.png to fig03_6.png and fig08.png.
- figures_2305.09710/ holds fig01, fig02, fig04, fig07 and fig08 (board Figures 22-26).
- 2108.08863.txt is the full text.

Blocked — Fix the contested evidence and fill the strong end of the simulation bracket. Obtain verbatim quotes for the key numbers in 1104.1174 (OWLS), 2410.17109 (FLAMINGO jet and low-fgas variants, S(k=1) and S_min) and 2605.04176 (Astrid). Use these to build a weak-to-strong bracket of S(k=1) and S_min at z=0..
(Two things are missing. First, numerical S(k=1) and S_min for FLAMINGO's fgas-8sigma and jet models appear only in plots (Figs. 5, 6, 11 and 13 of 2410.17109), never in the text; getting them means extracting those figures and reading values off the curves. Second, the tool budget ran out before cite() was called, so references.bib has no entries yet for 2410.17109 or 2605.04176, and the 1104.1174 quote is not re-cited. Dead end x-04 records this.) This direction is partly done. I quoted the three papers verbatim from their full texts and posted the results as f-018 to f-021. The strong end of the bracket is placed by the papers' own statements, but the text gives no numerical S(k=1) or S_min for FLAMINGO's strong models, so that success criterion is not met. All values are at z=0.

Bracket, weak to strong: (1) Astrid (arXiv:2605.04176, f-021) is the weak end. Abstract: 'no significant suppression at z = 0 and mild suppression at z = 0.2, weaker than in other state-of-the-art simulations'. Text: at z=0.2 mild suppression extends to k~1 h/Mpc, then the ratio 'rises back to above unity at z = 0'. So S(k=1) is about 1 or above, with no S_min below 1. The relevant figure is Fig. 1. (2) Horizon-AGN (arXiv:1801.08559, f-012 from d-02): less than 15% suppression at k~10 h/Mpc. That excerpt was not re-quoted here. (3) FLAMINGO fiducial (arXiv:2410.17109, f-019): the minimum is at k=10 h/Mpc and the ratio first departs from unity at k~0.1 h/Mpc. For the thermal gas-fraction variants the minimum stays at k=8-10 h/Mpc. (4) OWLS AGN (arXiv:1104.1174, f-018) is labelled intermediate. Abstract: suppression of 1% at k~0.3, 10% at k~1 and 30% at k~10 h/Mpc, so S(k=1)~0.90 and S(k=10)~0.70. (5) FLAMINGO strong-end variants (f-019, f-020): the jet model gives a larger response than the thermal model at the same gas-fraction setting. The thermal and jet versions of the model whose cluster gas fractions sit 8 sigma below the calibration data ('fgas-8sigma') 'closely resemble' cosmo-OWLS

with AGN heating temperature log $\Delta T = 8.3$ and 8.5. The thermal response stops growing below that setting. Both are 'within the ranges demanded' by Bigwood+2024 but 'in strong tension' with X-ray/WL gas fractions. The Amon & Efstathiou response is 'out of reach' of FLAMINGO. Only Illustris and cosmo-OWLS $\Delta T = 8.7$ lie beyond the emulator's range. The relevant figures are Figs. 2, 5, 6, 11 and 13 of 2410.17109.

Full texts are in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-baryonic-pk/runs/261017_110039/ (1104.1174.txt, 2410.17109.txt, 2605.04176.txt).

Critic, wave 2 — continue: 8 findings stand up to the evidence.

Issues: The task explicitly asks for emulator-predicted Pk-ratios at z=0 for a couple of parameter points. No emulator was run, and no S(k=1) or S(k=5) value exists for any named parameter point. The final report must state plainly that every S(k) value is a literature value. An analytic stand-in may still be possible: the SP(k) or van Daalen+2019 fitting formulae evaluated by hand from published coefficients. That route has not been tried.; The evidence excerpts show only the first few hundred characters of 10,000-16,000 character text chunks. None of the quoted claim strings is actually visible in the excerpts: the Amod = 0.69 +- 0.06 quote, the 21-28% band, the FLAMINGO 'closely resemble' sentence and the Astrid 'rises back to above unity' sentence. I accepted these findings because the chunk ranges plausibly cover the cited sections (abstracts at chars 0-4000, Sec. 4 later in the text) and the quotes are internally consistent with earlier findings. Future findings should store the exact matching snippet as the excerpt.; f-018 ranks OWLS AGN as 'stronger than FLAMINGO fiducial'. No quoted number supports this, because the FLAMINGO fiducial S(k=1) and S_min are not given in the text. The only support is the OWLS 10% at k~1. The ordering against Horizon-AGN is also loose: Horizon-AGN gives under 15% at k~10, while OWLS gives 30% at k~10. Treat the ranking as provisional..

Stopping after wave 2: budget reached (2/2 waves used). Writing the report from the board.

Research report:

Baryonic suppression of the matter power spectrum: literature survey, AGN feedback physics, and weak-lensing implications (no emulator evaluation was possible)

Abstract

The task had three parts: predict the baryonic power-spectrum ratio S(k) = P_hydro/P_DMO at z=0 for at least two feedback settings using a hydrodynamics emulator, explain the AGN-feedback physics of the suppression, and assess its impact on weak-lensing cosmology.

The emulator part could not be done. No emulator was available in this session, and none could be installed. No S(k) value in this report was computed by us. Every number is a published value, quoted from paper text.

Simulation bracket at z=0. The quoted values span:

- The weak end: Astrid shows no significant suppression; Horizon-AGN shows less than 15% at k~10 h/Mpc.
- An intermediate case: OWLS AGN shows 1% at k~0.3, 10% at k~1 and 30% at k~10 h/Mpc.
- The strong end: the FLAMINGO jet and low-gas-fraction variants resemble cosmo-OWLS with log ΔT_AGN = 8.3 to 8.5.

An X-ray-calibrated BCemu fit gives a maximum suppression of 21-28% (68% CL) at k~7 h/Mpc.

Physical mechanism. AGN feedback ejects gas from group-scale haloes (M ~ 1e13-1e14 Msun/h) to beyond r200m. The amplitude of the suppression is set by the mean baryon fraction of groups.

Weak lensing. Reconciling KiDS-1000 or DES Y3 cosmic shear with Planck ΛCDM through baryons alone requires:

- A_mod = 0.69 ± 0.06 for KiDS;
- A_mod = 0.858 ± 0.052 for DES Y3;
- a suppression that reaches down to k~0.2 h/Mpc.

This is stronger than most simulations produce. FLAMINGO concludes that baryonic effects cannot remove the S8 tension. Ignoring feedback shifts S8 by roughly 1σ in the quoted KiDS and DES analyses.

Background

OWLS reference. The OWLS result of van Daalen et al. (arXiv:1104.1174) established that AGN feedback suppresses the z=0 matter power by percent-to-tens-of-percent levels over the scales probed by cosmic shear. Radiative cooling raises the power only at very small scales.

Baryon-fraction relations. Two papers tie the suppression to halo baryon fractions:

- van Daalen et al. 2019 (arXiv:1906.00968) used a library of 92 OWLS, cosmo-OWLS and BAHAMAS power spectra. The mean baryon fraction in groups predicts the suppression to 1% for $k < 1$ h/Mpc.
- SP(k) (arXiv:2305.09710) extended this to a 400-run suite. It provides a public code (pyspk).

Parameterised and emulated models. The baryonic correction model (arXiv:1510.06034) expresses the mechanism in parametric form:

- the ejected gas fraction sets the amplitude;
- the ejection radius sets the scale where suppression begins.

This approach underlies the BCemu emulator (arXiv:2108.08863) and the BACCO emulator (arXiv:2011.15018). The FLAMINGO project has three relevant papers:

- a Gaussian-process emulator of its response (arXiv:2410.17109);
- an S8 analysis (arXiv:2309.07959);
- a study of how feedback couples to cosmology (arXiv:2403.12967).

Halo-mass decomposition. Two papers split the suppression by halo mass: arXiv:2309.06479 for cosmo-OWLS/BAHAMAS and arXiv:2511.10634 for IllustrisTNG.

Weak-suppression simulations. Horizon-AGN (arXiv:1801.08559) and Astrid (arXiv:2605.04176) predict weak suppression.

S8 analyses. The weak-lensing S8 question is addressed by:

- the "non-linear solution" papers (arXiv:2206.11794, arXiv:2305.09827);
- a DES Y3 + ACT kSZ joint fit (arXiv:2404.06098);
- KiDS × tSZ (arXiv:2109.04458);
- DES Y3 harmonic-space shear (arXiv:2203.07128);
- an early forecast of the dark-energy bias from neglecting baryons (arXiv:1105.1075).

Search conditions. The abstract-search tool was rate-limited (HTTP 429) on all three attempts. The survey was therefore built from a second search tool, followed by full-text downloads of the core papers.

Results

1. Emulator evaluation: not possible No tool in the session could evaluate P_hydro/P_DMO:

- The input-file and dataset lists were empty.
- Dispatch was local.
- The only job tool (run_test_job) draws random samples from a normal distribution. It does not evaluate any physics.

BCemu and pyspk could not be installed. Saved paper figures could not be used as a substitute either: the figure tool stores PNGs and captions, but no tool could view or digitise images.

Consequently:

- No named weak or strong parameter point (for example a low and a high BCemu log Mc) was evaluated.
- No S(k=1) or S(k=5) value exists for any named point.
- The SP(k) relation between f_b and S was not reproduced.

2. Literature S(k) at z=0: the simulation bracket All values below are published, not computed. In the table, k(1%) is the k at which suppression first reaches 1%.

Model	Ordering	Quoted z=0 behaviour	k(1%)	S at stated k	Minimum
Astrid (arXiv:2605.04176)	weak	"no significant suppression at z = 0"; mild suppression to k~1 h/Mpc at z=0.2, with the ratio rising back above unity at z=0	not quoted	S(k=1) about 1 or above	none below 1 quoted

Model	Ordering	Quoted z=0 behaviour	k(1%)	S at stated k	Minimum
Horizon-AGN (arXiv:1801.08559)	weak	less than 15% at k~10 h/Mpc; enhancement by a factor of a few at smaller scales; non-monotonic evolution over z=0-5	not quoted	S(10) > 0.85	not quoted
FLAMINGO fiducial (arXiv:2410.17109)	moderate	departs from unity at k~0.1 h/Mpc; minimum moves from k=6 h/Mpc (z=2) to k=10 h/Mpc (z=0)	~0.1 h/Mpc (departure)	not quoted	k_min = 10 h/Mpc
OWLS AGN (arXiv:1104.1174)	intermediate (provisional)	1% at k~0.3, 10% at k~1, 30% at k~10 h/Mpc	~0.3 h/Mpc	S(1)~0.90, S(10)~0.70	at or beyond 10 h/Mpc
FLAMINGO fgas-8σ thermal / jet	strong	"closely resemble" cosmo-OWLS log ΔT = 8.3 (thermal) and 8.5 (jet); jet response exceeds thermal at fixed fgas	not quoted	not quoted	k_min 8-10 h/Mpc for thermal fgas variants

Further details on the strong end, from the FLAMINGO paper:

- Below fgas-8σ the thermal response "reaches a saturation value".
- Only Illustris and cosmo-OWLS ΔT = 8.7 lie beyond the emulator's reach.
- The numerical amplitudes for these strong variants appear only in the paper's Figs. 5, 6, 11 and 13, which could not be read.

Astrid results on black-hole dynamics:

- Black-hole repositioning boosts kinetic AGN feedback, by a factor of 2 at z=1.5.
- At z=0 the repositioning run lies about 10% lower than the dynamical-friction run around k = 20 h/Mpc.
- Modelling choices for black-hole dynamics therefore shift the predicted ratio substantially.

Caveat on ordering: OWLS AGN is labelled stronger than FLAMINGO fiducial, but no quoted FLAMINGO-fiducial S(k=1) or S_min supports this. Its ordering relative to Horizon-AGN

is only partly constrained: 30% versus less than 15% at k~10 h/Mpc. The ranking should be treated as provisional. An earlier pass called OWLS the "strong/fiducial" reference; that label was revised to "intermediate".

3. X-ray-calibrated BCemu constraint Giri and Schneider (arXiv:2108.08863, Sec. 4.2) fit the 7-parameter BCemu model to X-ray group and cluster gas fractions. The fit assumes Planck cosmology and hydrostatic bias b_hse = 0.26 ± 0.07, and the posterior sits near log10 Mc ~ 12.6. The posterior prefers low δ: a slowly falling gas profile, with much of the gas ejected beyond the virial radius.

The resulting z=0 suppression:

- **68% CL:** suppression exceeds 1% between k~0.1 and 0.4 h/Mpc. It grows to a maximum of 21-28% (S_min ~0.72-0.79) at k~7 h/Mpc. The abstract rounds this to 20-28%.
- **99% CL:** k(1%) lies at 0.1-1.3 h/Mpc. The maximum is 14-38% (S_min ~0.62-0.86) beyond k~7 h/Mpc.

How the paper places simulations against this band:

- Only cosmo-OWLS 8.5 is fully consistent.
- cosmo-OWLS 8.0, BAHAMAS and OWLS lie between the 68% and 99% regions.
- Illustris-TNG and Horizon-AGN fall outside the 99% region, because their suppression is shifted to higher k.

The OWLS-AGN 30% at k~10 h/Mpc sits at or just beyond the strong edge of the 68% band. The FLAMINGO strongest variants are stated to be in strong tension with X-ray gas fractions.

BCemu coverage:

- Parameter ranges: log10 Mc ? [11, 15], θ_ej ? [2, 8], δ ? [3, 11], η ? [0.05, 0.4], f_b ? [0.10, 0.25].
- Wavenumber range: k ~ 0.03-12.5 h/Mpc.
- Emulation error: below 1% at 1σ.

[[FIGURE 1]]

Figure 1 is the BCemu emulator's percentage error on S(k) for its test set at several redshifts, with 1σ and 2σ bands. It documents the stated sub-percent emulation accuracy.

[[FIGURE 2]]

Figure 2 is the paper's comparison of S = P/P_N-body from the hydrodynamical simulations with best-fit baryonification curves at three redshifts. It was saved but not read numerically.

[[FIGURE 3]]

Figure 3 is one panel of the paper's Figure 3. It compares the baryonification model, fitted only to simulated gas and stellar fractions, with the S(k) measured in cosmo-OWLS, BAHAMAS, OWLS, Illustris-TNG and Horizon-AGN. The remaining panels are registered separately as Figures 4-8 and are not repeated here.

[[FIGURE 21]]

Figure 21 is the paper's Figure 8: the constraints on f_gas, f_star and S(k) from the X-ray observations. It is the figure behind the 21-28% band. No values were read from it; the numbers above come from the text.

4. Physics of the suppression The physical picture rests on abstract-level and introductory quotes from the papers below. The critic noted that the stored evidence excerpts for these items show only file headers, so they are not independently verified at the verbatim level.

Gas ejection. Feedback heats gas and ejects it to large distances, which is needed to match X-ray groups (arXiv:1906.00968). In haloes with M500c ? 1e13 Msun/h, AGN feedback moves matter from inside r500c to r200m and possibly beyond (arXiv:2309.06479).

Halo-mass dependence.

- For k ? 3 h/Mpc, group-sized haloes (1e14±0.25 Msun/h) dominate the power spectrum. At smaller scales, the combined contribution of lower-mass haloes takes over.
- A lower AGN heating temperature damps the effect at k ? 4 h/Mpc for M500c $\geq$ 1e13.25 (arXiv:2309.06479).
- In IllustrisTNG, haloes with log M200m in [13, 14] are reported to dominate at k~2-30 h/Mpc (arXiv:2511.10634). This comes from the survey stage only; the full text was not read.

Baryon fraction as the controlling variable.

- The mean group baryon fraction alone predicts the z=0 suppression to 1% for k $<$ 1 h/Mpc. Differences in group baryon fraction explain why simulations disagree (arXiv:1906.00968).
- A parameter-free model built on halo baryon fractions is accurate to the percent level down to k~10 h/Mpc (arXiv:2309.06479).

Cooling. Radiative cooling dominates only for k ? 70 h/Mpc in OWLS (arXiv:1104.1174). Horizon-AGN shows an enhancement of a factor of a few at scales smaller than k~10 h/Mpc (arXiv:1801.08559).

Cosmology coupling. Feedback strength depends on cosmology through f_b/(Ω_m σ8)^(1/4), which yields an accurate fitting formula. Combined baryonic and non-baryonic suppression (neutrinos, decaying dark matter) exceeds the sum of the two (arXiv:2403.12967).

The SP(k) figures shown below were saved but not read. The SP(k) full text was not read. No SP(k) result is drawn from them here.

[[FIGURE 23]]

Figure 23 is the SP(k) fractional baryonic impact on total matter power for all 400 models at z=0.125, colour-coded by the total baryon fraction of haloes of M200c = 1e14 Msun. It is the relation the task asked to reproduce.

[[FIGURE 25]]

Figure 25 is the SP(k) parameterisation of suppression as a function of the scaled baryon fraction (their Eq. 4), with the scale and redshift dependence of the fit parameters λ, μ and ν. These published coefficients are what a by-hand evaluation of SP(k) would use.

[[FIGURE 26]]

Figure 26 compares the SP(k) model with simulations for 8 randomly selected runs.

5. Impact on weak-lensing cosmology Required suppression in the A_mod model. Amon and Efstathiou (arXiv:2206.11794) model the power spectrum as P = P_L + A_mod(P_NL - P_L).

For KiDS-1000, with the cosmology fixed to Planck TTTEEE ΛCDM, the $\xi\pm$ fit gives "a mean Amod = 0.69 ± 0.06", with χ^2_min = 265.5 for 225 data points. The authors note that this error is underestimated because it neglects the Planck cosmology uncertainty.

The same fit done with HMCode2020 needs Θ_AGN = 9.17 (+0.2, -0.8). This is well outside the BAHAMAS-calibrated range of 7.6-8.0. The paper calls the required suppression "more extreme than predicted by most of the hydrodynamic simulations".

The tension being addressed, against Planck S8 = 0.829 ± 0.012:

- KiDS shear S8 = 0.759 (+0.024, -0.021): about 9% (2.4-2.7σ) low.
- DES Y3 shear S8 = 0.772 (+0.018, -0.017): about 7% (2.3-2.6σ) low.

DES Y3 refit. Preston, Amon and Efstathiou (arXiv:2305.09827) give, with a Planck prior:

Data	Scale cuts	A_mod	Significance
DES Y3	none	0.858 ± 0.052	2.7σ from 1
KiDS, same prior	—	0.748 ± 0.072	—
DES Y3 + KiDS (naive combination)	—	0.820 ± 0.042	suppression required at about 4σ

For comparison, BAHAMAS with log ΔT_heat = 8.0 "is well approximated by Amod = 0.9" for k ? 1 h/Mpc. The data require more suppression than that, closer to the C-OWLS 8.5/8.7 runs, and it must extend to k~0.2 h/Mpc. The paper names non-standard dark matter as an alternative explanation.

S8 shift when feedback is ignored.

- KiDS (arXiv:2206.11794, Table 1): S8 = 0.755 ± 0.016 with no feedback, against 0.774 ± 0.021 with the fiducial Θ_AGN 7.3-8.3 prior. The shift is about 0.02, roughly 1σ.
- DES Y3 (arXiv:2305.09827, Table 2):
- All scales, no feedback: S8 = 0.793 ± 0.012, which is 1.8σ below Planck (0.828 ± 0.016).
- With scale cuts: 0.805 ± 0.017, which is 0.9σ below Planck.
- With A_mod and free cosmology: 0.831 ± 0.036.

The DES difference of about 0.04 compares fits that differ in both the model and the treatment of cosmological freedom. It should therefore be read as an upper-bound-style estimate, not a controlled shift.

FLAMINGO verdict (arXiv:2309.07959). FLAMINGO compares its full suite, including extreme variants, with cosmic shear, CMB lensing, tSZ and their cross-spectra. Its conclusions:

- Baryonic effects are not large enough to remove the S8 tension.
- Current shear data are only marginally sensitive to baryons.
- Stronger feedback improves the fit to KiDS/DES in the D3A cosmology only slightly, and this preference largely disappears once a lower S8 is adopted.

- CMB lensing agrees well with Planck ΛCDM.
- The strongest tension comes from tSZ y-y, then shear × tSZ, then shear-shear.

This conclusion rests on abstract-level reading plus a body excerpt about the ordering of tensions. No verbatim conclusion quote was captured.

A companion FLAMINGO statement (arXiv:2410.17109) supports this. The Amon and Efstathiou response, and to a lesser extent the Preston et al. response, "are out of reach of our emulator and likely of what can be achieved using the FLAMINGO model without altering the sub-grid models".

WL + kSZ (arXiv:2404.06098). The DES Y3 WL + ACT DR5 kSZ fit uses BCEmu baryonification. Results:

- S8 = 0.823 (+0.019, -0.020), higher than WL alone.
- The fit prefers a mean feedback scenario more extreme than most simulations.
- The baryon fraction at M500 ~ 1e13